

Problem 1: Externalities - Taxation

A company emits pollutants into the air, causing a negative externality. The marginal cost of production for the company is given by $MC=50+0.5q$. The marginal social cost, accounting for the pollution's externalities, is $MSC=50+1.5q$. The market price is constant at \$75.

- Determine the socially efficient quantity of production.
- Given the company's marginal cost function, what quantity would the company produce in the absence of regulation?
- Calculate the Pigovian tax rate necessary to achieve the socially efficient quantity.

Problem 2: Positive Externalities - Subsidies

A local government is considering subsidizing a public transportation system to encourage use and reduce traffic congestion, which has positive externalities. The marginal social benefit (MSB) of the transportation system is given by $MSB=200-2q$, where q represents the number of users. The marginal private benefit (MPB), representing user demand, is $MPB=150-2q$. The marginal cost of operating the system is constant at 40.

- Determine the socially optimal number of users (socially efficient quantity).
- What quantity will occur without any subsidy?
- Calculate the subsidy per user required to achieve the socially optimal quantity.

Problem 3:

- 1) Consider the market for college education in the United States. College degrees have become quite expensive in the U.S., but around 1 million people in the U.S. earn a degree each year. Suppose the yearly supply and demand functions for degrees in the U.S. are:

$$S(p) = 20p$$
$$D(p) = 1,200,000 - 10p$$

- Using these supply and demand functions, find the equilibrium price and quantity in the private market for 4-year degrees.
- Suppose each person with a college degree creates an external benefit for society that can be valued at \$10,000. Graph the private supply and demand curves, and label the equilibrium that you found in part a. Then, add to your graph the *social demand curve* for college degrees.
- Label the section of your graph showing the *deadweight loss* from the externality of a college degree. Is this DWL coming from over- or underproduction?
- If the government were to try to correct this market failure, would it be better to use a tax to do so, or a subsidy? What is the correct value for the tax or subsidy they should use?
- When taking the external benefits of the externality into account, the social demand function for college degrees is:

$$D(p) = 1,300,000 - 10p$$

Using this new demand curve, calculate the socially optimal price and quantity for a 4-year college degree.

- If the government does not intervene in the market for college degrees, what is the dollar value of the deadweight loss in the private equilibrium?